1 $5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3}$ **[M1]**

$4\sqrt{3}$ **[A1]**

[Total Marks: 2]

2 $5\sqrt{3}$ or $4\sqrt{3}$ or $3\sqrt{3}$ seen **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 2]

3 $6\sqrt{2}$ or $5\sqrt{2}$ or $3\sqrt{2}$ seen **[M1]**

$8\sqrt{2}$ **[A1]**

[Total Marks: 2]

4 $5\sqrt{3}$ or $3\sqrt{3}$ or $2\sqrt{3}$ seen **[M1]**

$4\sqrt{3}$ **[A1]**

[Total Marks: 2]

5 $\sqrt{25 \times 3} + \sqrt{16 \times 3} - \sqrt{9 \times 3} = 5\sqrt{3} + 4\sqrt{3} - 3\sqrt{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 2]

6 (a) $3\sqrt{5}$ **[B1]**

(b) $2\sqrt{5}$ or $4\sqrt{5}$ seen **[M1]**

$6\sqrt{5}$ **[A1]**

[Total Marks: 3]

7 $\sqrt{75} = 5\sqrt{3}$ **[B1]**

$\sqrt{27} = 3\sqrt{3}$ or $\sqrt{12} = 2\sqrt{3}$ **[B1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 3]

8 $\sqrt{48} = 4\sqrt{3}$ **[B1]**

$\sqrt{12} = 2\sqrt{3}$ **[B1]**

$4\sqrt{3} - 2\sqrt{3} = 2\sqrt{3}$ **[B1]**

[Total Marks: 3]

9 **(a)** $\sqrt{25 \times 3}$ or $\sqrt{25}\sqrt{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

(b) $6\sqrt{2} - 2\sqrt{2}$ **[M1]**

$4\sqrt{2}$ **[A1]**

[Total Marks: 4]

10 **(a)** **(i)** $3\sqrt{5} + 2\sqrt{5}$ **[M1]**

$5\sqrt{5}$ **[A1]**

(ii) Multiplies numerator and denominator by $\sqrt{3}$: $\dfrac{18\sqrt{3}}{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 4]

11 **(a)** $3\sqrt{5}$ **[B1]**

(b) $6\sqrt{2} + 5\sqrt{2} - 3\sqrt{2}$ **[M1]**

$8\sqrt{2}$ **[A1]**

(c) $\dfrac{21\sqrt{7}}{\sqrt{7}\sqrt{7}}$ or $\dfrac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Total Marks: 5]

12 **(a)** $8 + 4\sqrt{3} - 2\sqrt{3} - 3$ **[M1]**

$5 + 2\sqrt{3}$ **[A1]**

(b) $\dfrac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Total Marks: 4]

13 **(a)** $9 - 3\sqrt{5} - 3\sqrt{5} + 5$ (or $9 - 6\sqrt{5} + 5$) **[M1]**

$14 - 6\sqrt{5}$ **[A1]**

(b) $\dfrac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

14 **(a)** $12 + 3\sqrt{5} - 8\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) Multiplies numerator and denominator by $\sqrt{7}$ to get $\frac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

15 (a) (a) $12 + 3\sqrt{5} - 8\sqrt{5} - 2(5)$ or $12 - 5\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) $\frac{15 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$ or $\frac{15\sqrt{3}}{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

[Total Marks: 4]

16 (a) $12 - 3\sqrt{2} + 4\sqrt{2} - 2$ **[M1]**

$10 + \sqrt{2}$ **[A1]**

(b) Multiply numerator and denominator by $(5 + \sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Total Marks: 5]

17 (a) $20 - 9$ or $4(5) - 9$ **[M1]**

11 **[A1]**

(b) Multiplies numerator and denominator by $(3 + \sqrt{2})$ **[M1]**

Denominator of $9 - 2$ or 7 seen **[M1]**

$6 + 2\sqrt{2}$ **[A1]**

[Total Marks: 5]

18 (a) $12 + 15\sqrt{2} - 2\sqrt{2} - 5$ **[M1]**

$7 + 13\sqrt{2}$ **[A1]**

(b) Multiplies numerator and denominator by $(4 + \sqrt{3})$ **[M1]**

Denominator of $16 - 3$ or 13 **[M1]**

$8 + 2\sqrt{3}$ or $2(4 + \sqrt{3})$ **[A1]**

[Total Marks: 5]

19 (a) $\frac{15\sqrt{5}}{5}$ **[M1]**

$3\sqrt{5}$ **[A1]**

(b) $10 - 2\sqrt{3} + 5\sqrt{3} - 3$ **[M1]**

$7 + 3\sqrt{3}$ **[A2]**

[Total Marks: 5]

20 **(a)** $6 - 3\sqrt{5} + 2\sqrt{5} - 5$ **[M1]**

$1 - \sqrt{5}$ **[A1]**

(b) $\dfrac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$\dfrac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 5]

21 **(a)** $15 + 5\sqrt{7} - 6\sqrt{7} - 2(7)$ **[M1]**

$1 - \sqrt{7}$ **[A1]**

(b) Multiplies numerator and denominator by $(5 + \sqrt{3})$: $\dfrac{22(5 + \sqrt{3})}{(5 - \sqrt{3})(5 + \sqrt{3})}$ **[M1]**

Denominator becomes $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Total Marks: 5]

22 **(a)** $20 + 8\sqrt{2} - 15\sqrt{2} - 6(2)$ **[M1]**

$8 - 7\sqrt{2}$ **[A1]**

(b) $\dfrac{11(4 + \sqrt{5})}{(4 - \sqrt{5})(4 + \sqrt{5})}$ **[M1]**

$\dfrac{11(4 + \sqrt{5})}{16 - 5}$ or $\dfrac{11(4 + \sqrt{5})}{11}$ **[M1]**

$4 + \sqrt{5}$ **[A1]**

(c) $3x\sqrt{3} - x\sqrt{3} = 18$ **[M1]**

$2x\sqrt{3} = 18$ or $x\sqrt{3} = 9$ **[M1]**

$x = \dfrac{9}{\sqrt{3}} = 3\sqrt{3}$ **[A1]**

[Total Marks: 8]

23 $\times \dfrac{3 + \sqrt{7}}{3 + \sqrt{7}}$ **[M1]**

Numerator $15 + 5\sqrt{7} + 3\sqrt{7} + 7$ or $22 + 8\sqrt{7}$ **[B1]**

Denominator $9 - 7$ or 2 **[B1]**

$11 + 4\sqrt{7}$ **[A1]**

[Total Marks: 4]

24 Forms an equation for the area: $(a + b\sqrt{3})(2 - \sqrt{3}) = 1 + 4\sqrt{3}$ **[M1]**

Expands the brackets: $2a - a\sqrt{3} + 2b\sqrt{3} - 3b = 1 + 4\sqrt{3}$ **[M1]**

Equates rational and irrational parts to form simultaneous equations: $2a - 3b = 1$ and $-a + 2b = 4$ **[M1]**

Attempts to solve the simultaneous equations (e.g. multiplies second equation by 2: $-2a + 4b = 8$ and adds to first to get $b = 9$) **[M1]**

$a = 14$, $b = 9$ **[A1]**

[Total Marks: 5]

25 (a) Multiplies numerator and denominator by $(2 + \sqrt{3})$ **[M1]**

Numerator expands to $10 + 5\sqrt{3} + 2\sqrt{3} + 3$ **[M1]**

Denominator is $4 - 3 = 1$ leading to final answer $13 + 7\sqrt{3}$ **[A1]**

(b) $x = \dfrac{14}{3 - \sqrt{2}}$ **[M1]**

Multiplies numerator and denominator by $(3 + \sqrt{2})$ to get $\dfrac{14(3 + \sqrt{2})}{9 - 2}$ **[M1]**

$x = 6 + 2\sqrt{2}$ **[A1]**

[Total Marks: 6]

26 (a) Multiply numerator and denominator by $2 + \sqrt{3}$ **[M1]**

Numerator becomes $(5 + \sqrt{3})(2 + \sqrt{3}) = 10 + 5\sqrt{3} + 2\sqrt{3} + 3$ (giving $13 + 7\sqrt{3}$) **[M1]**

$13 + 7\sqrt{3}$ **[A1]**

(b) Length $= \dfrac{12 + 4\sqrt{5}}{3 + \sqrt{5}}$ **[M1]**

Multiply numerator and denominator by $3 - \sqrt{5}$ **[M1]**

Numerator becomes $(12 + 4\sqrt{5})(3 - \sqrt{5}) = 36 - 12\sqrt{5} + 12\sqrt{5} - 20 = 16$ **[M1]**

4 **[A1]**

[Total Marks: 7]

27 (a) Multiplies numerator and denominator by $(5 + \sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3$ or 22 **[M1]**

$5 + \sqrt{3}$ **[A1]**

(b) $x(\sqrt{2}-1)=\sqrt{8}$ or $x(\sqrt{2}-1)=2\sqrt{2}$ **[M1]**

$x=\dfrac{2\sqrt{2}}{\sqrt{2}-1}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{2}+1)$ **[M1]**

$4+2\sqrt{2}$ **[A1]**

[Total Marks: 7]

28 (a) $\dfrac{1}{2}\times(2+\sqrt{3})\times h=5+3\sqrt{3}$ **[M1]**

$h=\dfrac{10+6\sqrt{3}}{2+\sqrt{3}}$ **[M1]**

Multiply numerator and denominator by $(2-\sqrt{3})$ **[M1]**

$h=\dfrac{20-10\sqrt{3}+12\sqrt{3}-18}{4-3}=2+2\sqrt{3}$ **[A1]**

(b) $y^2=(2+\sqrt{3})^2+(2+2\sqrt{3})^2$ **[M1]**

$y^2=(4+4\sqrt{3}+3)+(4+8\sqrt{3}+12)$ **[M1]**

$23+12\sqrt{3}$ **[A1]**

[Total Marks: 7]

29 (a) $\dfrac{5+2\sqrt{3}}{2+\sqrt{3}}$ **[M1]**

Multiplies numerator and denominator by $(2-\sqrt{3})$ **[M1]**

$4-\sqrt{3}$ **[A1]**

(b) $x\sqrt{5}-2x=2+\sqrt{20}$ or $x(\sqrt{5}-2)=\ldots$ **[M1]**

$x=\dfrac{2+2\sqrt{5}}{\sqrt{5}-2}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{5}+2)$ **[M1]**

$14+6\sqrt{5}$ **[A1]**

[Total Marks: 7]

30 (a) $\dfrac{14-2\sqrt{5}}{3+\sqrt{5}}$ **[M1]**

Multiplies numerator and denominator by $(3-\sqrt{5})$ **[M1]**

Numerator expands to $42-14\sqrt{5}-6\sqrt{5}+10$ (or $52-20\sqrt{5}$) **[M1]**

$13-5\sqrt{5}$ **[A1]**

(b) Uses Pythagoras' theorem: $(3+\sqrt{5})^2+(13-5\sqrt{5})^2$ **[M1]**

$14+6\sqrt{5}$ or $294-130\sqrt{5}$ seen **[M1]**

$308-124\sqrt{5}$ with no errors seen **[A1]**

[Total Marks: 7]

31 **(a)** Multiplies numerator and denominator by $(3+\sqrt{3})$: $\dfrac{6(3+\sqrt{3})}{(3-\sqrt{3})(3+\sqrt{3})}$ **[M1]**

Expands denominator correctly: $9-3$ or 6 **[M1]**

$\dfrac{6(3+\sqrt{3})}{6}$ **[M1]**

$3+\sqrt{3}$ **[A1]**

(b) $\dfrac{2\sqrt{6}+3\sqrt{6}}{\sqrt{3}}$ or $\dfrac{5\sqrt{6}}{\sqrt{3}}$ or $5\sqrt{2}$ **[M1]**

$\dfrac{10\sqrt{2}}{2}$ or $5\sqrt{2}$ **[M1]**

0 (or $0\sqrt{2}+0\sqrt{3}$) **[A1]**

[Total Marks: 7]

32 **(a)** Area $= \dfrac{1}{2} \times$ base $\times$ height $\implies 5+2\sqrt{2} = \dfrac{1}{2} \times (3+\sqrt{2}) \times h$ **[M1]**

$h = \dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ **[A1]**

(b) Multiply numerator and denominator by $(3-\sqrt{2})$: $\dfrac{2(5+2\sqrt{2})(3-\sqrt{2})}{(3+\sqrt{2})(3-\sqrt{2})}$ **[M1]**

Denominator: $3^2-(\sqrt{2})^2 = 9-2 = 7$ **[B1]**

Numerator: $2(15-5\sqrt{2}+6\sqrt{2}-4) = 2(11+\sqrt{2}) = 22+2\sqrt{2}$ **[M1]**

Height $= \dfrac{22+2\sqrt{2}}{7}$ **[A1]**

(c) $\dfrac{1(\sqrt{7}+2)}{(\sqrt{7}-2)(\sqrt{7}+2)}$ **[M1]**

$\dfrac{\sqrt{7}+2}{7-4} = \dfrac{\sqrt{7}+2}{3}$ **[A1]**

[Total Marks: 8]

33 **(a)** $\dfrac{1}{2} \times (\sqrt{6}-\sqrt{2})(\sqrt{6}+\sqrt{2})$ **[M1]**

$\dfrac{1}{2}(6-2)$ **[M1]**

2 **[A1]**

(b) $(\text{Hypotenuse})^2 = (\sqrt{6}-\sqrt{2})^2 + (\sqrt{6}+\sqrt{2})^2$ **[M1]**

$(6-2\sqrt{12}+2)+(6+2\sqrt{12}+2)$ or $8-2\sqrt{12}+8+2\sqrt{12}$ **[M1]**

$(\text{Hypotenuse})^2 = 16$ **[A1]**

$\text{Hypotenuse} = \sqrt{16} = 4$ **[A1]**

(c) Area scale factor $= \frac{18}{2} = 9$ **[M1]**

Length scale factor $= \sqrt{9} = 3$ **[M1]**

$4 \times 3 = 12$ **[A1]**

[Total Marks: 10]